SRE Lab Task: Data Flow Diagram (DFD)

Scenario: University Course Registration System

A university wants to develop an online Course Registration System through which students can view available courses and register for their desired courses.

Students provide their login credentials to access the system. After successful authentication, they can request a list of available courses and submit their selected course details for registration.

The system retrieves student information from the Student Records database and course information, including prerequisites and available seats, from the Course Records database. Before accepting a registration request, the system verifies whether the student has satisfied the course prerequisites and whether seats are available.

If all conditions are satisfied, the system records the registration in the Registration Records database and sends a registration confirmation to the student. Otherwise, the system rejects the request and provides an appropriate reason, such as unmet prerequisites or no available seats.

The university's academic administration staff can request registration reports from the system to review student registrations and course enrollment statistics.

Task: Level 1 DFD

- Decompose the main system into logical processes.
- Identify the required data stores.
- Connect processes, data stores, and external entities using appropriately labelled data-flow arrows.
- Ensure that every data flow represents data being transferred rather than an action or command.